

Posts and Telecommunications Institute of Technology
Campus at Hồ Chí Minh City

Semester 1, 2026-2027

Exercises Chapter 4

Student: <N24DECE104> , <Thái Tuấn Vũ>

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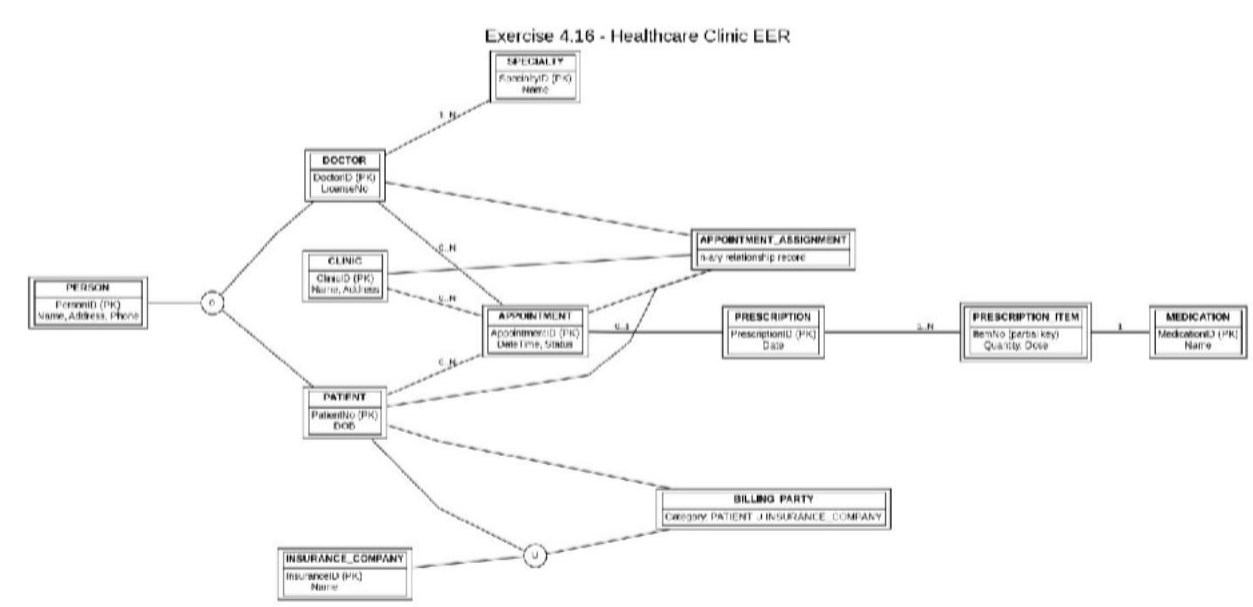
Exercise 4.16

Healthcare Clinic & Telemedicine Portal.

PERSON -> PATIENT/DOCTOR is overlapping and partial. PRESCRIPTION_ITEM is weak under PRESCRIPTION. BILLING_PARTY is a category over PATIENT INSURANCE_COMPANY.

APPOINTMENT_ASSIGNMENT is an n-ary relationship among PATIENT, DOCTOR and CLINIC.

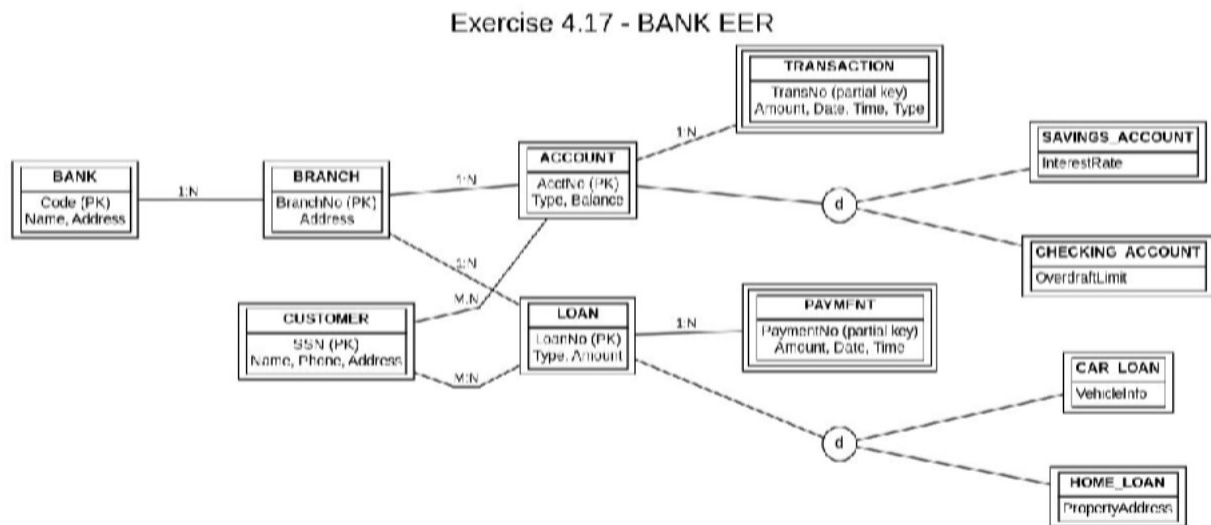
Each appointment has one patient, doctor and clinic; prescriptions contain one or more items; each item references one medication.



Exercise 4.17

BANK specialization/generalization.

The supplied 7th-edition text has the BANK schema as Figure 3.22, although Exercise 4.17 says Figure 3.21. ACCOUNT -> SAVINGS_ACCOUNT/CHECKING_ACCOUNT and LOAN -> CAR_LOAN/HOME_LOAN/OTHER_LOAN, assumed disjoint and total. Add TRANSACTION under ACCOUNT and PAYMENT under LOAN, storing amount/date/time.



Exercise 4.18

Olympic facilities.

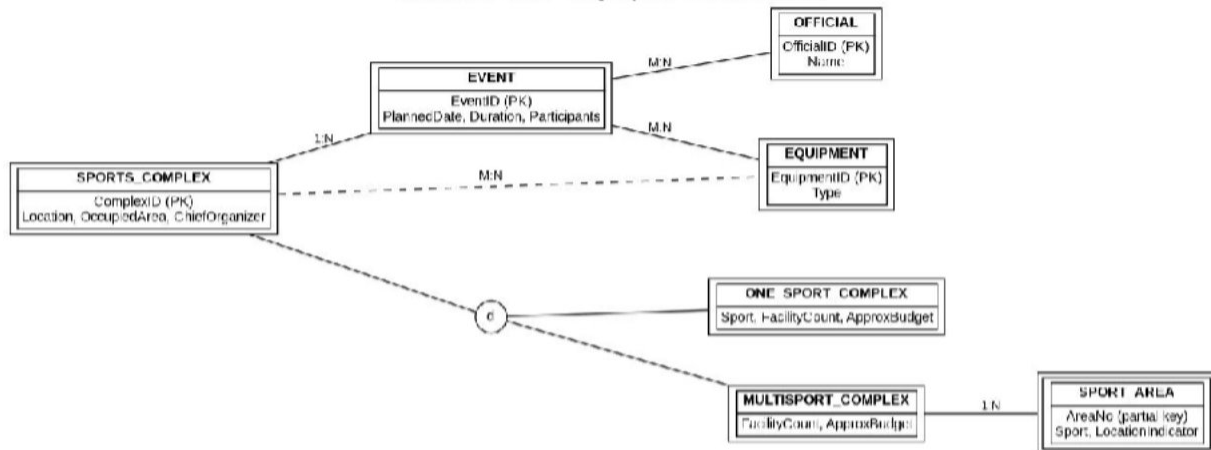
SPORTS_COMPLEX -> ONE_SPORT_COMPLEX/MULTISPORT_COMPLEX, disjoint and total.

MULTISPORT_COMPLEX owns weak SPORT_AREA records. EVENT is hosted by a complex;

OFFICIAL participates M:N; EQUIPMENT is linked to events and maintenance. Subclasses

store facility count and approximate budget.

Exercise 4.18 - Olympic Facilities EER

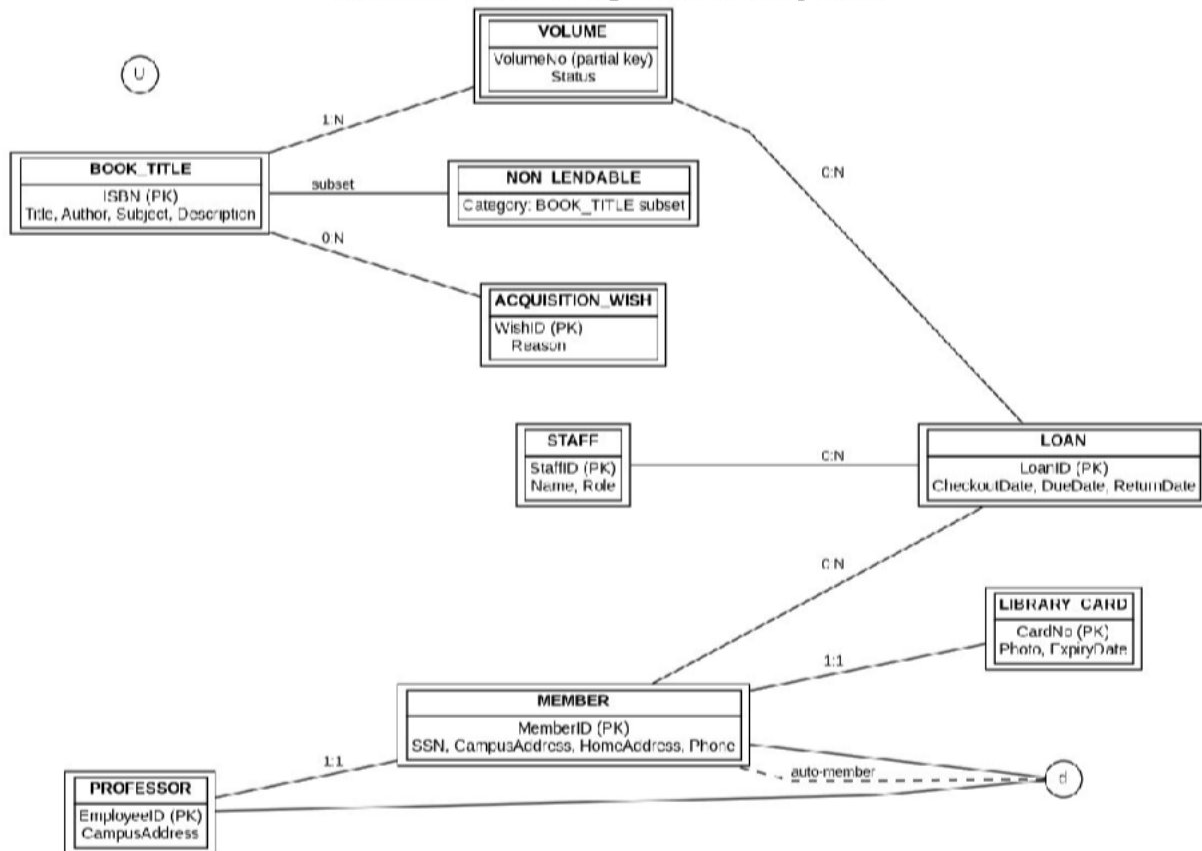


Exercise 4.19

Georgia Tech Library.

Use MEMBER, LIBRARY_CARD, BOOK_TITLE, VOLUME, LOAN and STAFF. VOLUME is weak under BOOK_TITLE; ISBN is the identifier because title is not unique. Members have at most five active books. Model professor automatic membership, non-lendable books and acquisition wishes; keep reminder/overdue rules as semantic constraints.

Exercise 4.19 - Georgia Tech Library EER

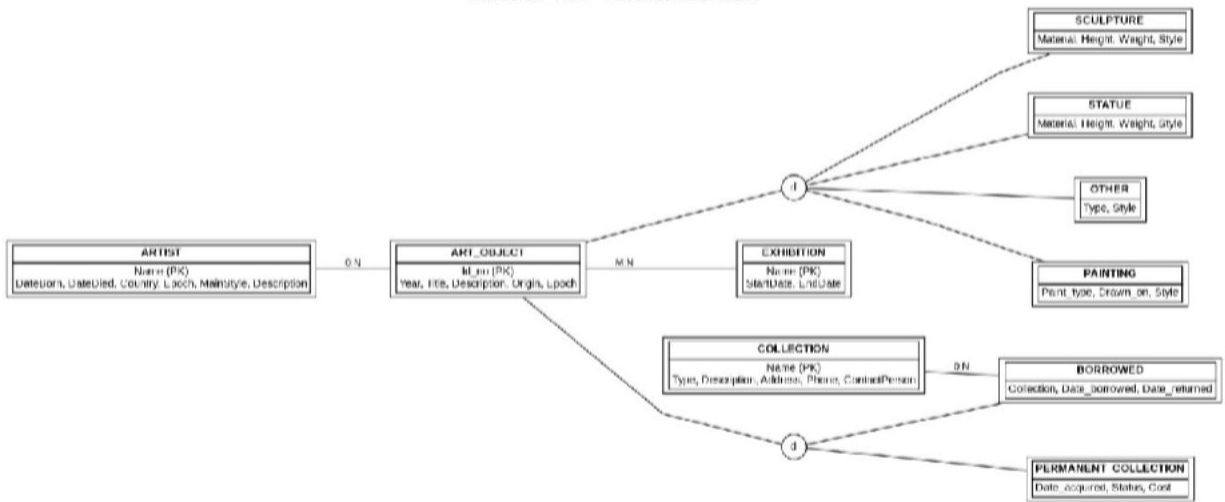


Exercise 4.20

Art museum.

ART_OBJECT -> PAINTING/SCULPTURE/STATUE/OTHER, disjoint and total; independently ART_OBJECT -> PERMANENT_COLLECTION/BORROWED, disjoint and total. ARTIST, EXHIBITION and COLLECTION are separate entities. EXHIBITION is M:N with ART_OBJECT.

Exercise 4.20 - Art Museum EER

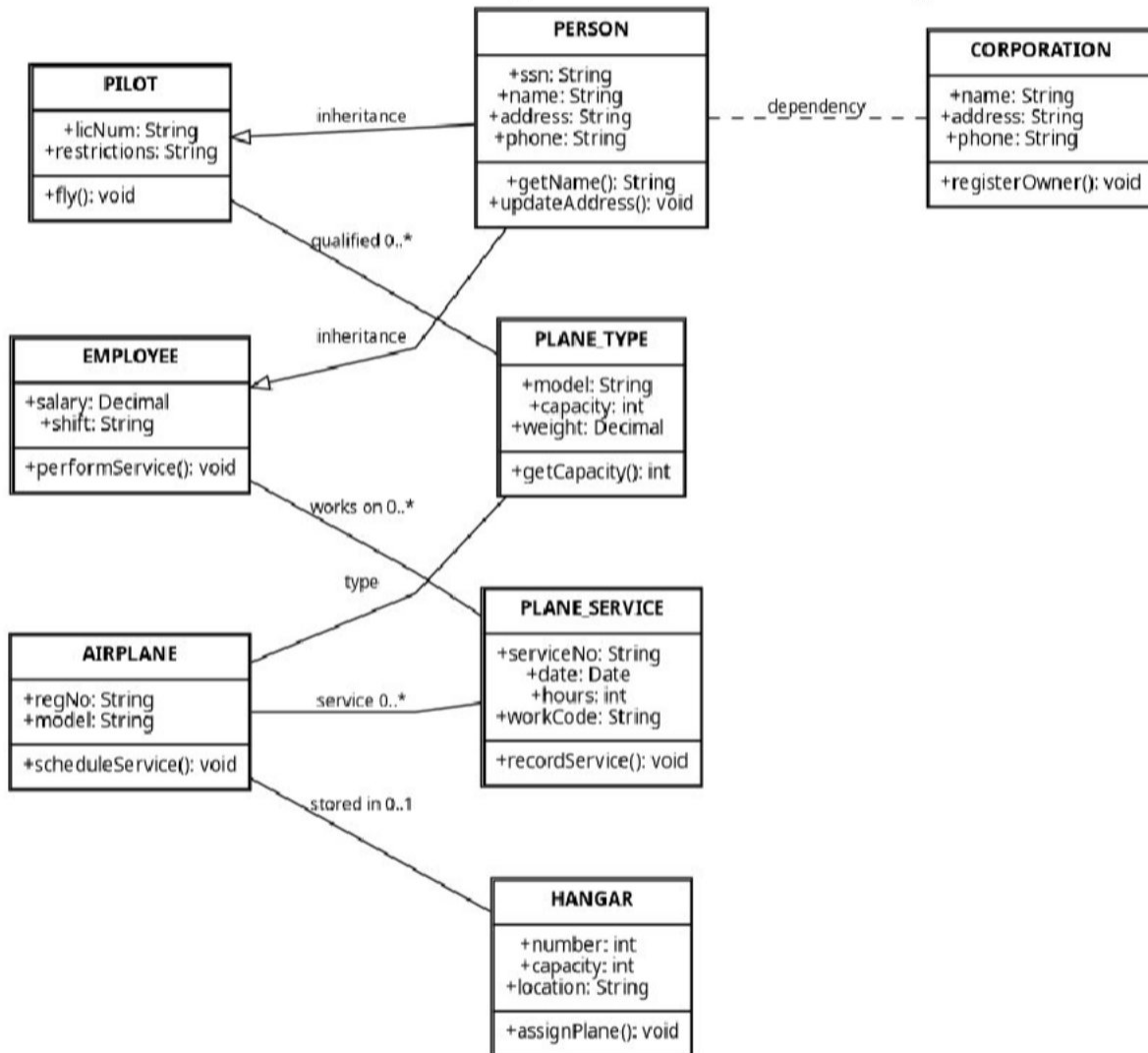


Exercise 4.21

SMALL_AIRPORT UML.

Every class is now a three-compartment UML class. Attributes and operations use + public visibility. PERSON -> PILOT and PERSON -> EMPLOYEE use hollow-triangle inheritance. Solid associations and a dashed dependency are explicitly shown.

Exercise 4.21 - SMALL_AIRPORT UML Class Diagram

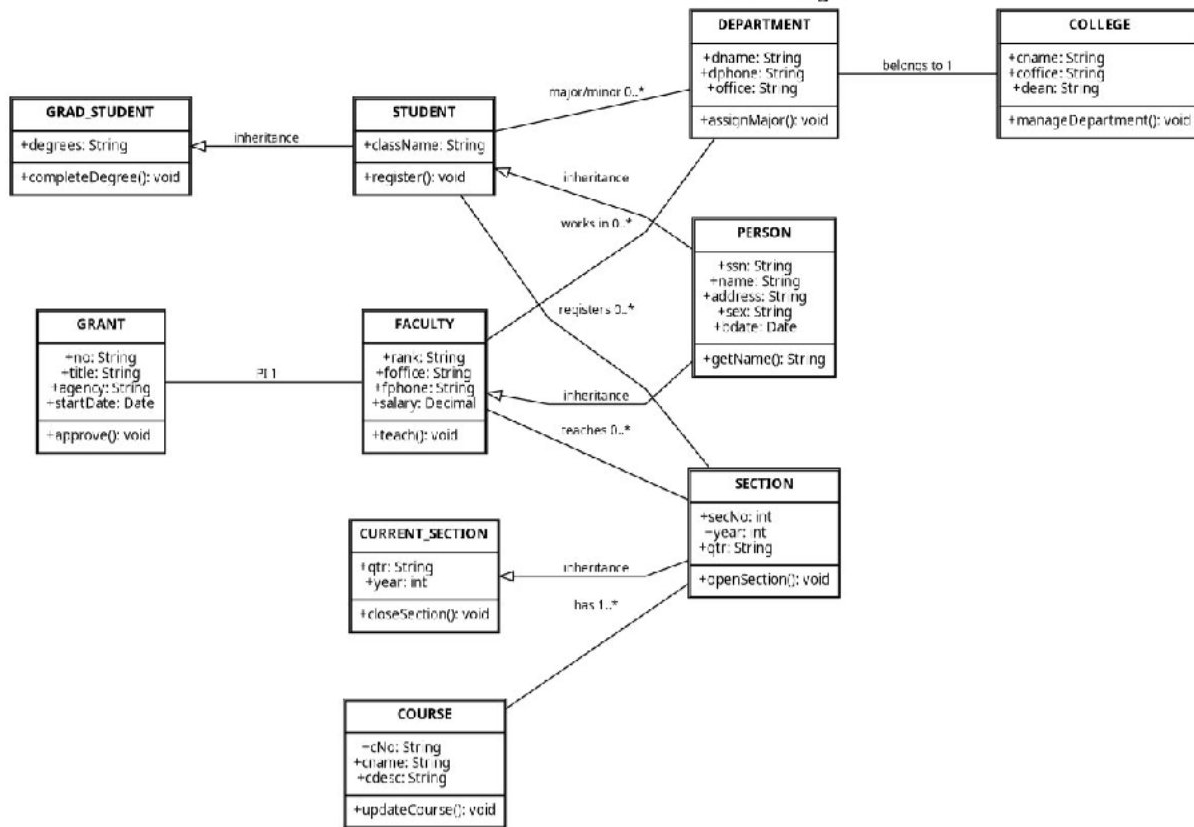


Exercise 4.22

UNIVERSITY EER in UML.

Use classes and generalization for PERSON -> FACULTY/STUDENT, STUDENT -> GRAD_STUDENT and SECTION -> CURRENT_SECTION. Use associations for BELONGS, MAJOR, MINOR, REGISTERED, TRANSCRIPT, TEACH, PI, SUPPORT, ADVISOR and COMMITTEE; preserve relationship attributes.

Exercise 4.22 - UNIVERSITY UML Class Diagram



Exercise 4.23

Classification table.

The exercise requires one checkmark in exactly one of four columns. The complete table is included immediately below.

STT	Entity Set / Attribute	(a) Relationship	(b) Attribute	(c) Specialization	(d) Generalization
1	MOTHER — PERSON	✓			
2	DAUGHTER — MOTHER	✓			
3	STUDENT — PERSON			✓	
4	STUDENT — Student_id		✓		
5	SCHOOL — STUDENT	✓			
6	SCHOOL —	✓			

	CLASS_ROOM				
7	ANIMAL — HORSE				✓
8	HORSE — Breed		✓		
9	HORSE — Age		✓		
10	EMPLOYEE — SSN		✓		
11	FURNITURE — CHAIR				✓
12	CHAIR — Weight		✓		
13	HUMAN — WOMAN				✓
14	SOLDIER — PERSON			✓	
15	ENEMY_COMBATANT — PERSON			✓	

Exercise 4.24

Draw a UML diagram to store information about a game of chess. State any assumptions made.

Assumptions

A chess game is played between exactly two players.

The game is played on an 8×8 board.

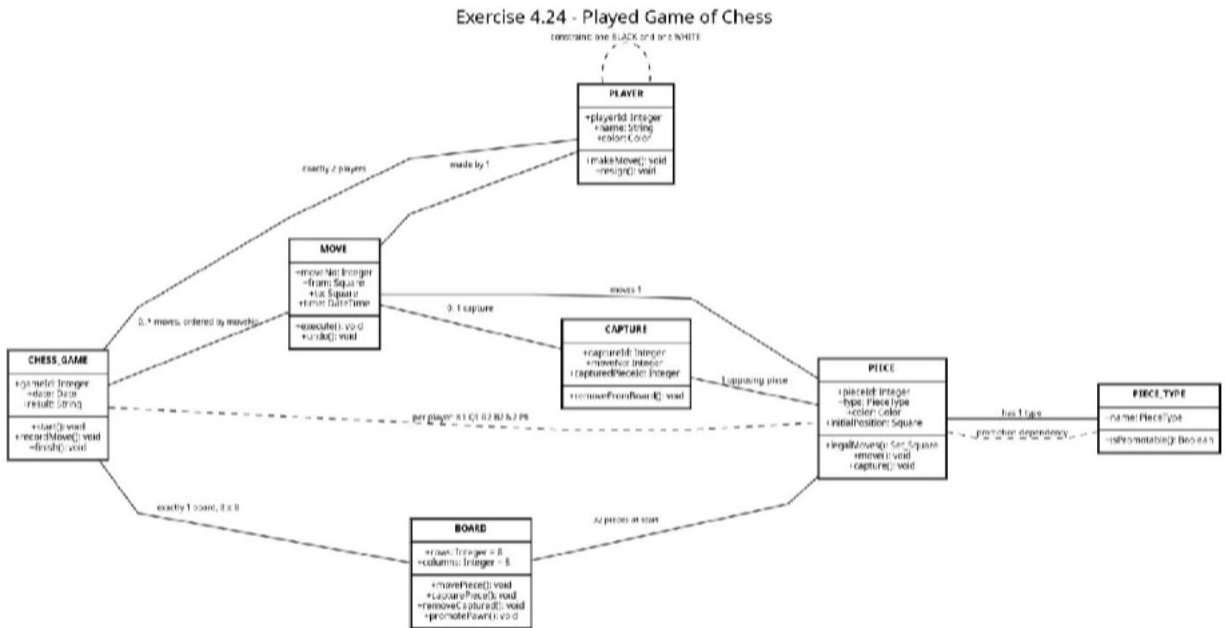
The two players are assigned the colors black and white.

Each player starts with one king, one queen, two rooks, two bishops, two knights, and eight pawns.

Every piece has a position on the board at the beginning of the game.

A player can move a piece only according to its legal moves. A move can capture an opposing piece; the captured piece is removed from the board.

When a pawn reaches the last rank, it can be promoted to a queen, rook, bishop, or knight.



Class descriptions

CHES_GAME: Stores the game ID, date and result, and records the sequence of moves.

PLAYER: Stores player information and the assigned color.

BOARD: Represents the 8×8 board and operations for moving, capturing, removing captured pieces and promoting pawns.

PIECE: Stores the piece type, color and initial position and provides legal-move, move and capture operations.

MOVE: Stores move number, source square, destination square and time.

CAPTURE: Stores the captured piece and the move that caused the capture; the captured piece is removed from the board.

PIECE_TYPE: Stores the type of chess piece and whether it is promotable.

Initial chess pieces

For each player: King $\times 1$, Queen $\times 1$, Rook $\times 2$, Bishop $\times 2$, Knight $\times 2$, Pawn $\times 8$. Therefore, each player has 16 pieces and the game begins with 32 pieces.

Move, capture and promotion

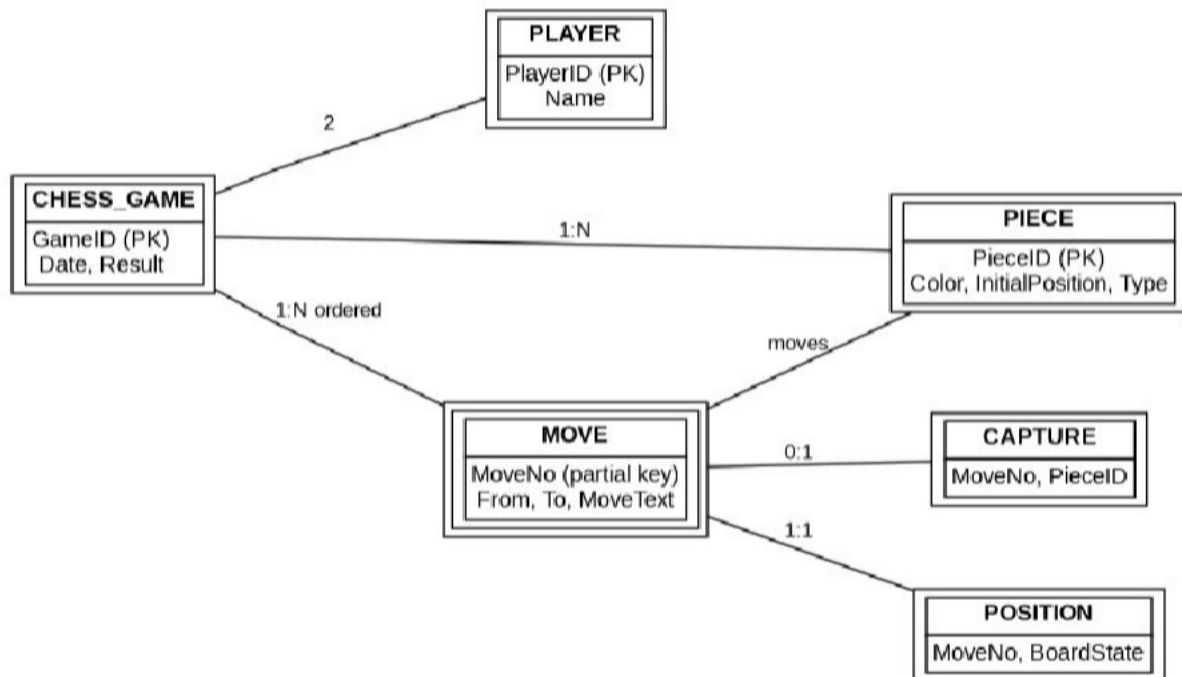
Each move records its order, source square, destination square and time. Legal moves are generated for the selected piece. If the destination contains an opposing piece, a capture is recorded and the captured piece is removed from the board. When a pawn reaches the last rank, it may be promoted to a queen, rook, bishop or knight.

Exercise 4.25

Chess persistent EER.

CHESS_GAME owns an ordered weak MOVE collection identified by MoveNo; PLAYER stores participants; PIECE stores persistent state; POSITION can store board state; CAPTURE records optional captured pieces.

Exercise 4.25 - Chess Persistent EER



Exercise 4.26

Which EER diagrams are incorrect?

None is inherently incorrect under the standard EER assumptions used here. The three original diagrams are reproduced below with their o/d specialization symbols and relationship cardinalities.

Diagram	Result	Reason	Symbols
(a)	CORRECT	Overlapping	o +

		specialization is allowed; E2 can independently participate in R with E3 (1:N).	double line
(b)	CORRECT	Disjoint E1/E2 may still participate in an independent 1:1 relationship R.	d + 1:1
(c)	CORRECT	Overlapping E1/E3 may participate in an independent M:N relationship R.	o + M:N

Conclusion: (a), (b), and (c) are not inherently invalid under the stated assumptions.

Exercise 4.27

The EER is kept aligned with the supplied figure: COMPUTER/LAPTOP/DESKTOP; ACCESSORY/KEYBOARD/MOUSE/MONITOR; COMPONENT/MEMORY/VIDEO_CARD/SOUND_CARD; SOFTWARE; OPERATING_SYSTEM; and SOLD_WITH, INSTALLED, INSTALLED_OS, SUPPORTS, OPTIONS, MEM_OPTIONS.

a) Attributes and keys

Entity	Key	Attributes	Reason
COMPUTER	ComputerID	Brand, Model, SerialNo, PurchaseDate	Unique computer.
LAPTOP	ComputerID	BatteryLife, Weight	Inherited supertype key.
DESKTOP	ComputerID	FormFactor, PowerRating	Inherited supertype key.
SOFTWARE	SoftwareID	Name, Version, LicenseType	Unique software record.
OPERATING_SYSTEM	OSID	Name, Version, Vendor	Unique OS record.
COMPONENT	ComponentID	SerialNo, Manufacturer, Warranty	Unique component.
MEMORY	ComponentID	CapacityGB, SpeedMHz	Inherited component key.
VIDEO_CARD	ComponentID	Chipset, VRAMGB	Inherited component key.
SOUND_CARD	ComponentID	Channels, AudioCodec	Inherited component key.
ACCESSORY	AccessoryID	Name, Manufacturer, Price	Unique accessory.
KEYBOARD	AccessoryID	Layout, ConnectionType	Inherited accessory

			key.
MOUSE	AccessoryID	DPI, ConnectionType	Inherited accessory key.
MONITOR	AccessoryID	SizeInches, Resolution	Inherited accessory key.

b) Maximum cardinalities and justification

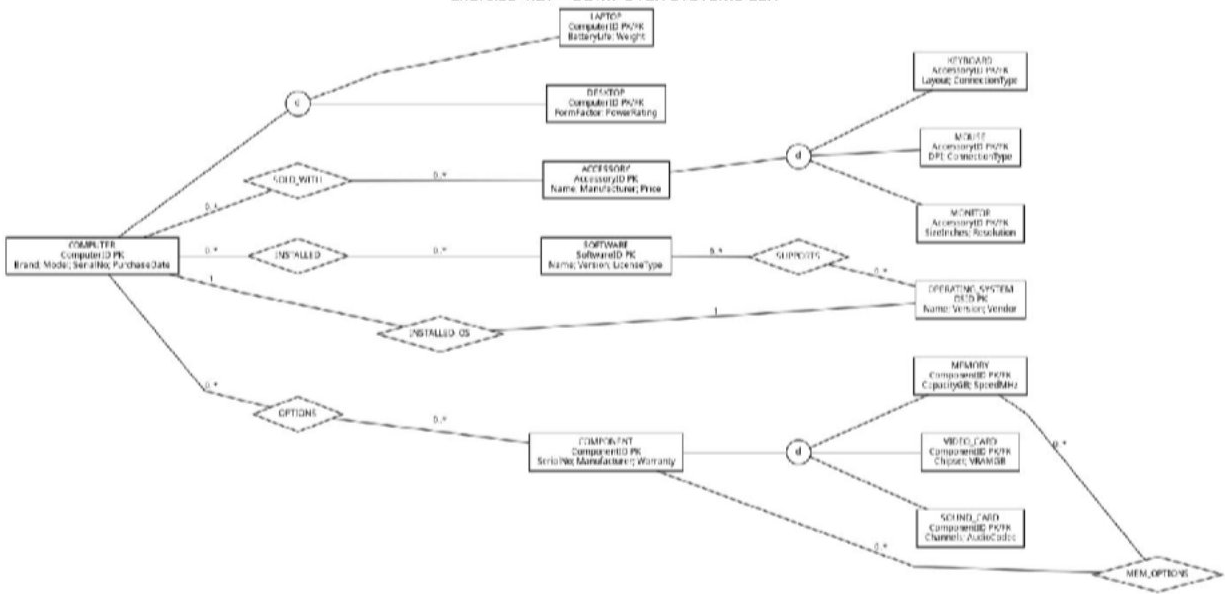
Relationship	Maximum	Justification
COMPUTER — INSTALLED — SOFTWARE	N:N	A computer can have many software packages; a software package can be installed on many computers.
COMPUTER — INSTALLED_OS — OPERATING_SYSTEM	N:1	Each computer has one installed OS in this simplified model; one OS can serve many computers.
SOFTWARE — SUPPORTS — OPERATING_SYSTEM	N:N	Software may support several OSs; an OS can support many software products.
COMPUTER — SOLD_WITH — ACCESSORY	N:N	A computer can be sold with several accessories; an accessory can be sold with many computers.
COMPUTER — OPTIONS — COMPONENT	N:N	A computer can have several component options; a component can be an option for many computers.
MEMORY — MEM_OPTIONS —	N:N	Memory options can be available for multiple compatible

COMPONENT		component/configuration records.
COMPUTER specialization	1:1	The supplied symbol means a computer belongs to one disjoint subtype: LAPTOP or DESKTOP.
ACCESSORY specialization	1:1	Each accessory record belongs to one of KEYBOARD, MOUSE or MONITOR.
COMPONENT specialization	1:1	Each component record belongs to one of MEMORY, VIDEO_CARD or SOUND_CARD.

c) Complete narrative description

The EER diagram describes the computer systems maintained by a company. COMPUTER is a supertype specialized into LAPTOP and DESKTOP. ACCESSORY is specialized into KEYBOARD, MOUSE and MONITOR. COMPONENT is specialized into MEMORY, VIDEO_CARD and SOUND_CARD. A computer can have software installed through INSTALLED and an operating system through INSTALLED_OS. SUPPORTS records software/operating-system compatibility. SOLD_WITH records accessories sold with a computer. OPTIONS records component choices for a computer, while MEM_OPTIONS records available memory options. The maximum cardinalities above make the intended participation of each entity explicit.

Exercise 4.27 - COMPUTER SYSTEMS EER

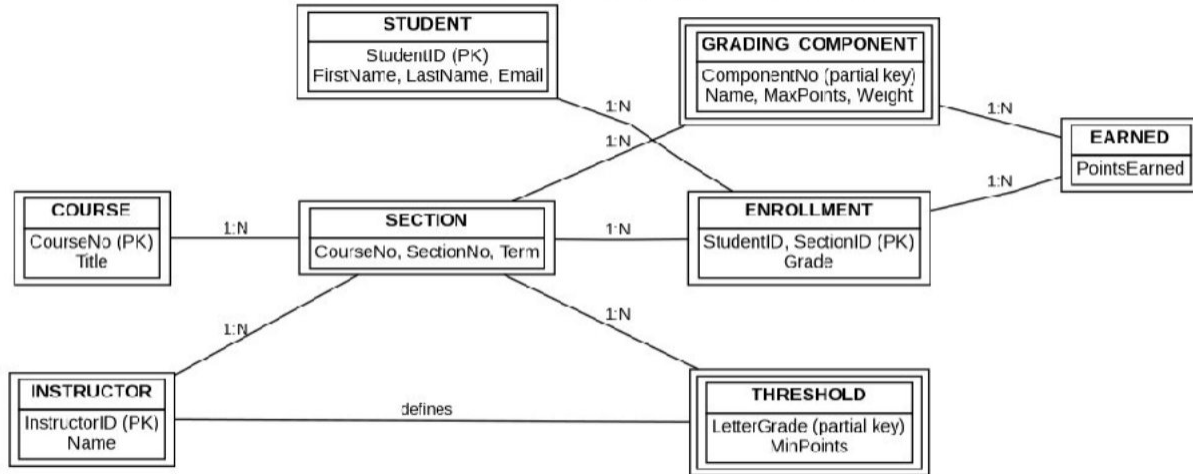


Exercise 4.28

GRADE_BOOK EER.

Entities: STUDENT, INSTRUCTOR, COURSE, SECTION, GRADING_COMPONENT, THRESHOLD and ENROLLMENT. Students enroll in sections; instructors teach sections; sections have grading components with maximum points/weights and A-F thresholds; earned points connect enrollments to components.

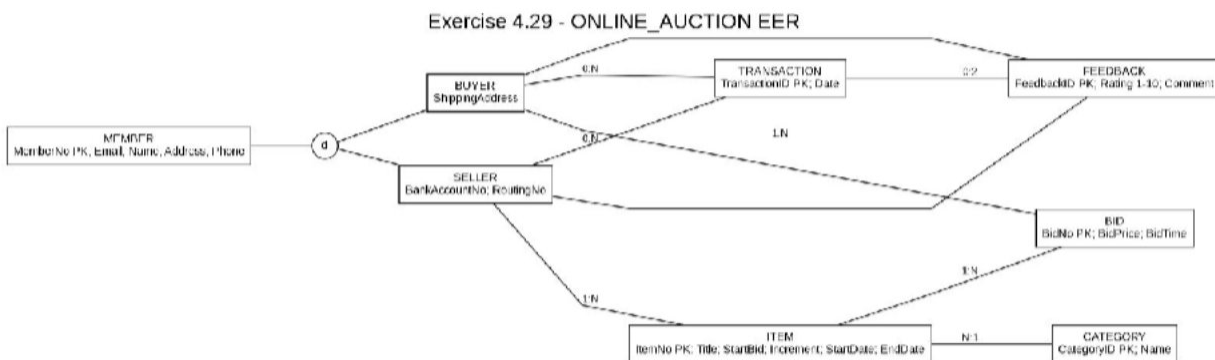
Exercise 4.28 - GRADE_BOOK EER



Exercise 4.29

ONLINE_AUCTION EER.

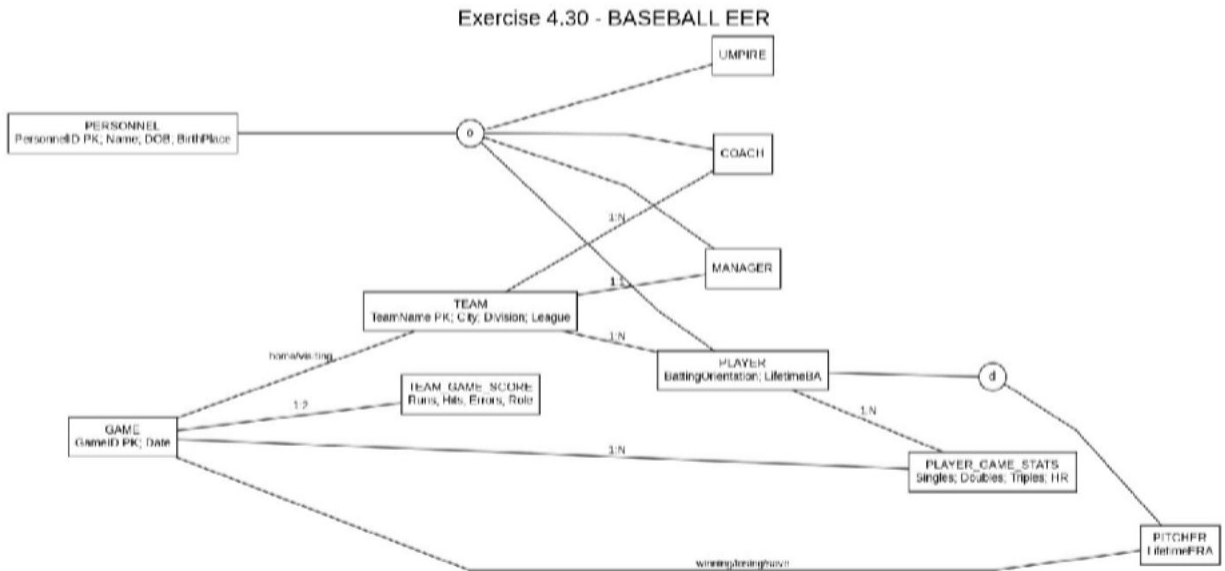
MEMBER -> BUYER/SELLER (assumed disjoint/total). SELLER places ITEMS; BUYER makes BIDS; ITEM belongs to a classification hierarchy; highest bid wins; completed sales form TRANSACTION; feedback is rated 1-10 with a comment.



Exercise 4.30

BASEBALL EER.

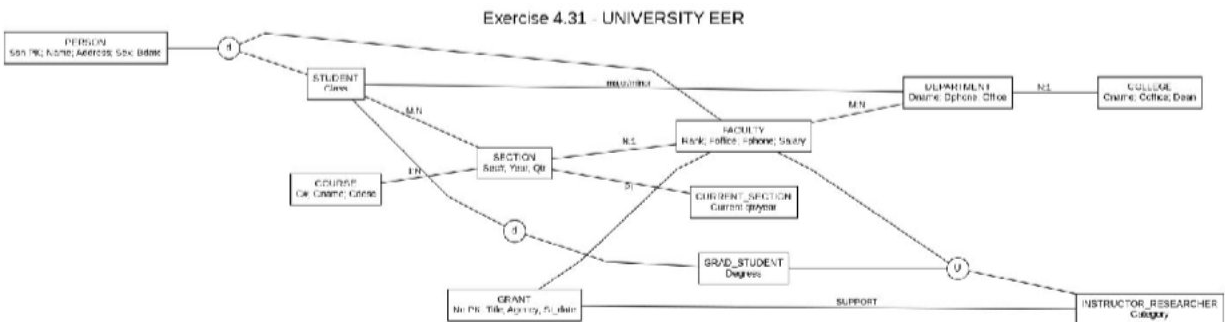
PERSONNEL -> PLAYER/COACH/MANAGER/UMPIRE; PLAYER -> PITCHER. TEAM has one manager, coaches and players. GAME has home/visiting teams, scores and winning/losing/ save pitcher roles; PLAYER_GAME_STATS stores singles, doubles, triples and home runs.



Exercise 4.31

UNIVERSITY EER equivalent notation.

Preserve Figure 4.9 semantics: PERSON/FACULTY/STUDENT/GRAD_STUDENT, CURRENT_SECTION, INSTRUCTOR_RESEARCHER category, DEPARTMENT/COLLEGE/COURSE/SECTION/GRANT and their relationships. A tool may use different glyphs while preserving semantics.

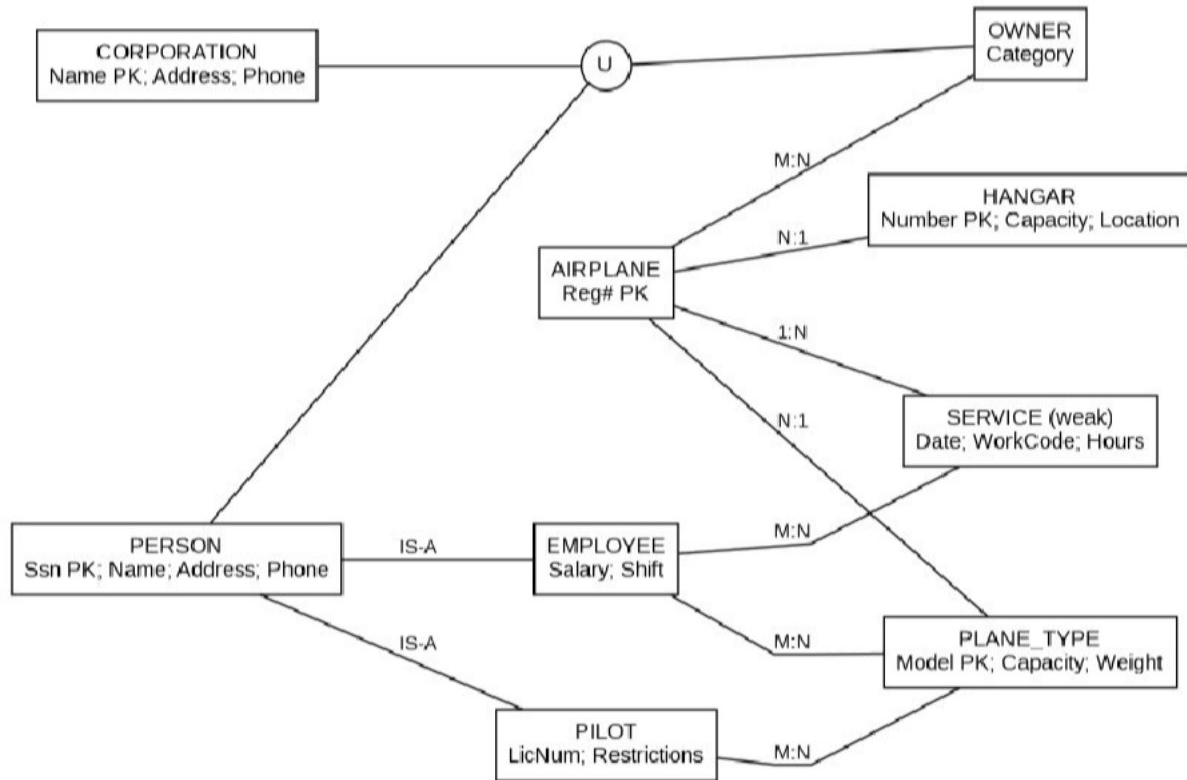


Exercise 4.32

SMALL_AIRPORT EER and OWNER category.

OWNER is a category subset of PERSON CORPORATION. Where union-category notation is unavailable, use PERSON_IS_OWNER and CORPORATION_IS_OWNER as separate relationships, matching the exercise hint. Preserve SERVICE as weak and PERSON -> PILOT /EMPLOYEE.

Exercise 4.32 - SMALL_AIRPORT EER

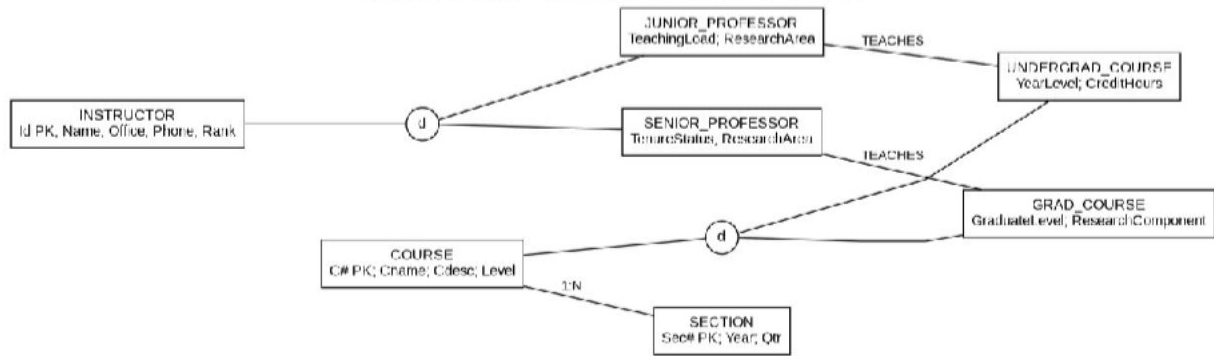


Exercise 4.33

Modified UNIVERSITY.

Create disjoint specializations INSTRUCTOR -> JUNIOR_PROFESSOR/SENIOR_PROFESSOR and COURSE -> UNDERGRAD_COURSE/GRAD_COURSE. Suggested local attributes: junior—TeachingLoad/ResearchArea; senior—TenureStatus/ResearchArea; undergraduate—YearLevel/CreditHours; graduate—GraduateLevel/ResearchComponent. Add TEACHES junior->undergraduate and senior->graduate.

Exercise 4.33 - Modified UNIVERSITY EER



End